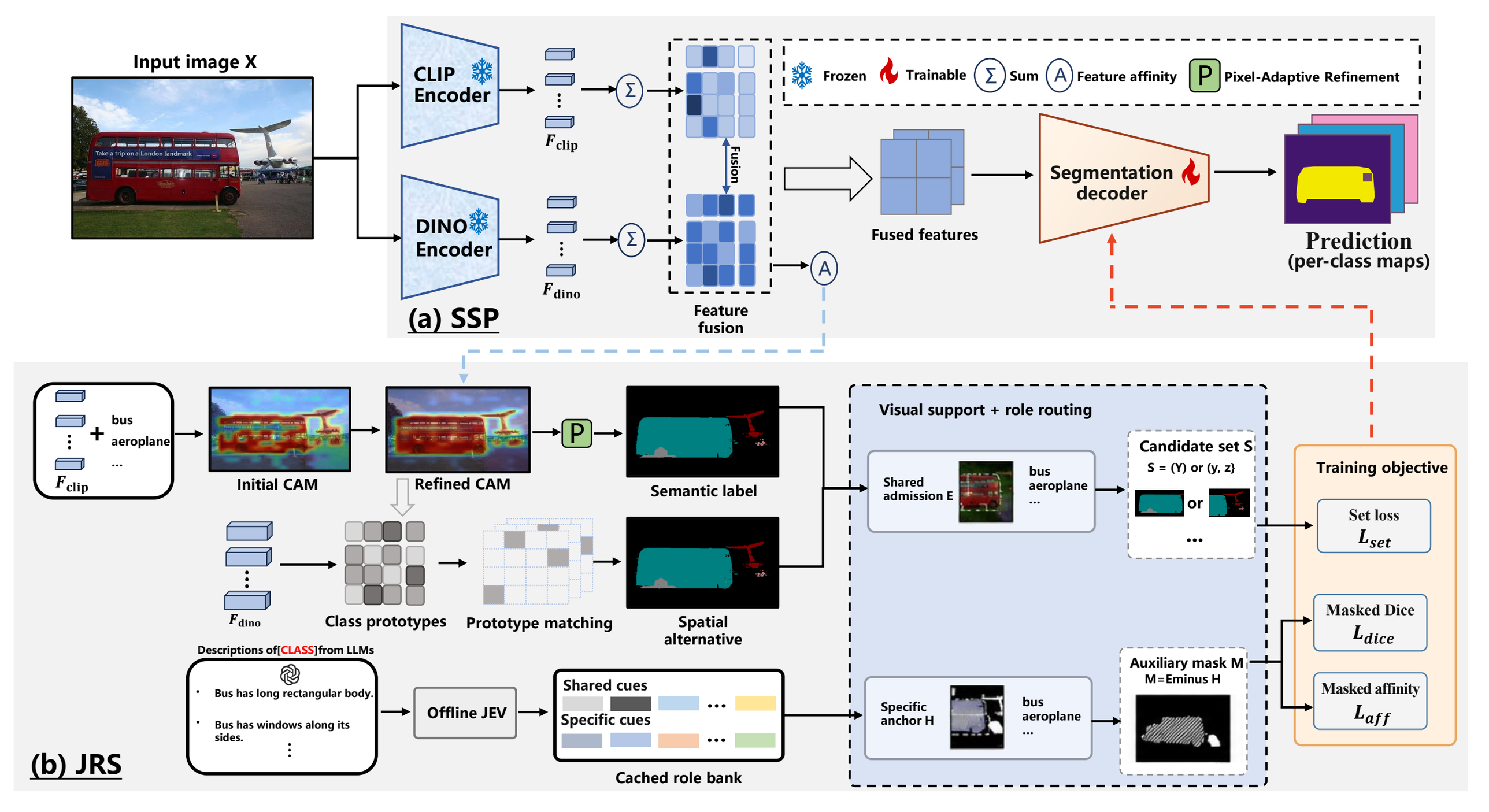

Input image X
CLIP Encoder
DINO Encoder
F_{clip}
F_{dino}
Fusion
Feature fusion
Frozen
Trainable
Sum
Feature affinity
Pixel-Adaptive Refinement
Fused features
Segmentation decoder
Prediction
(per-class maps)
(a) SSP
bus
aeroplane
...
Initial CAM
Refined CAM
Semantic label
Class prototypes
Prototype matching
Spatial alternative
Descriptions of[CLASS]from LLMs
Bus has long rectangular body.
Bus has windows along its sides.
Offline JEV
Shared cues
Specific cues
Cached role bank
Visual support + role routing
Shared admission E
Candidate set S
S = (Y) or (y, z}
or
Specific anchor H
Auxiliary mask M
M=Eminus H
Training objective
Set loss
L_{set}
Masked Dice
L_{dice}
Masked affinity
L_{aff}
(b) JRS